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9.2.1 Singular matrices and the eigenvalue problem

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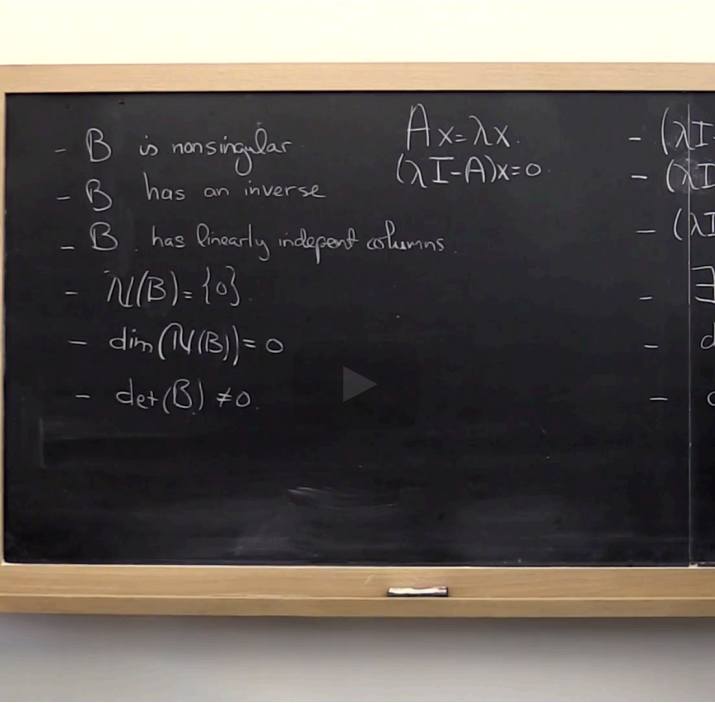
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space that only contains the zero vector, it's dimension of the null space

is equal to zero and the determinant B is not equal to zero. And as a matter of fact,

there was a whole list of these kinds of equivalent statements. And it turns out

that as we talk about eigenvalue problems, eigenvalues and eigenvectors,

what we'll do is we'll strategically pick the equivalent statement that is most

appropriate for a situation. Okay? Now

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Reading assignment

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Read Unit 9.2.1 of the notes. [\[LINK\]](#)

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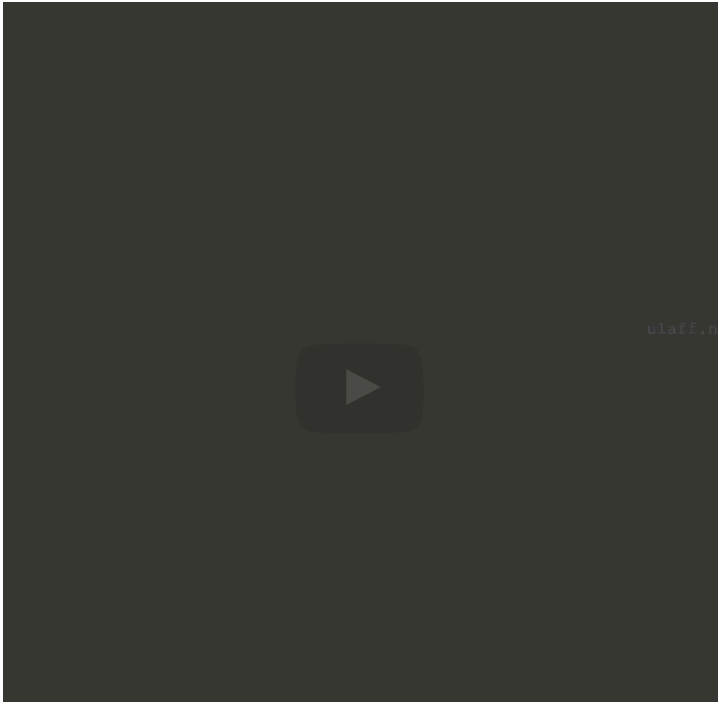
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matrix A. Okay. And that's known as the spectrum of

matrix A. The spectral radius of matrix A is defined as the largest absolute value

of any of the eigenvalues of matrix A. Okay?

So why do we call this the spectral radius. Well, you can think of all of the

eigenvalues, they're scalars, and therefore they are complex valued numbers, well

real or complex values. And if you put them all in the complex plane, and you



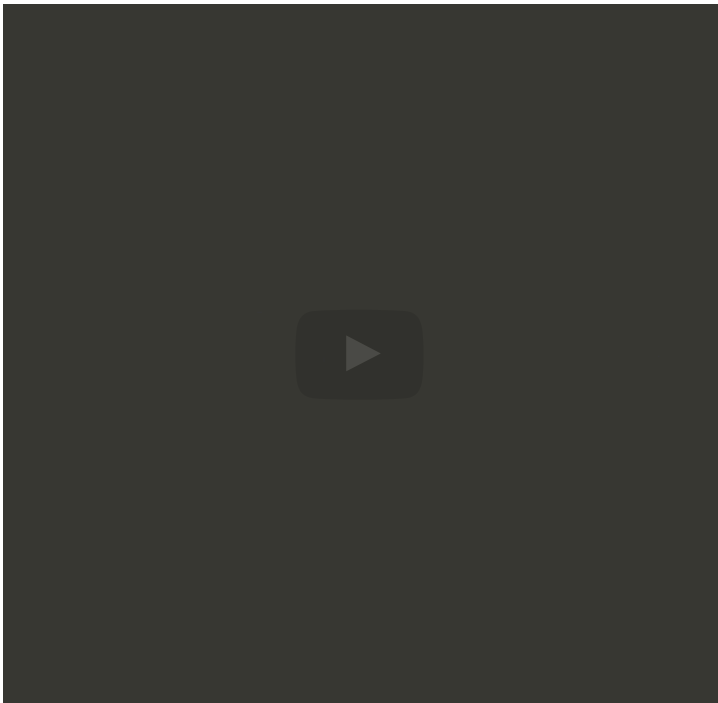
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diagonal elements, well the

absolute values of the off diagonal elements in the row indexed with 0. If you

now go to the point α_0 in the complex plane, and you draw a circle

around that with radius ρ_0 defined as such. And you do that for all of the

diagonal elements and the corresponding radii -

- so here we have another disk of

radius ρ_1 and another disk of radius ρ_2 and another disk of radius ρ_3 .



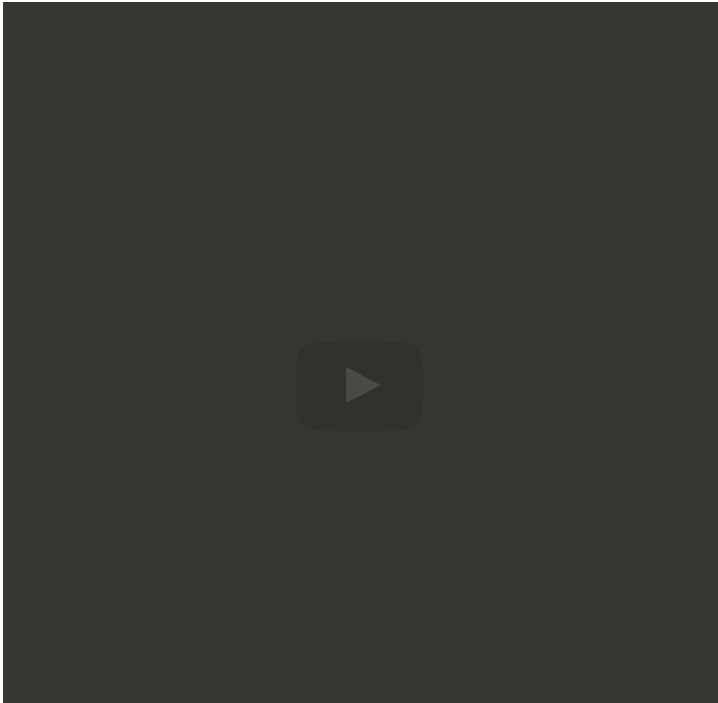
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Homework 9.2.1.1

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$.

TRUE/FALSE: $0 \in \Lambda(A)$ if and only A is singular.

TRUE ▼

✔ Answer: TRUE

Solution

For a complete solution, see the [notes](#).

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Homework 9.2.1.2

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$ be Hermitian.

ALWAYS/SOMETIMES/NEVER: All eigenvalues of A are real-valued.

ALWAYS ▼

✔ Answer: ALWAYS

Solution

For a complete solution, see the [notes](#).

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Homework 9.2.1.3

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$ be Hermitian positive definite (HPD).

ALWAYS/SOMETIMES/NEVER: All eigenvalues of A are positive.

ALWAYS   Answer: ALWAYS

Solution

For a complete solution, see the [notes](#).

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Homework 9.2.1.4

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$ be Hermitian, (λ, x) and (μ, y) be eigenpairs associated with A , and $\lambda \neq \mu$.

ALWAYS/SOMETIMES/NEVER: $x^H y = 0$

ALWAYS   Answer: ALWAYS

Solution

For a complete solution, see the [notes](#).

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Homework 9.2.1.5

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$, (λ, x) and (μ, y) be eigenpairs, and $\lambda \neq \mu$. Prove that x and y are linearly independent.

 done



Solution

For a complete solution, see the [notes](#).

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Homework 9.2.1.6

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$, $k \leq m$, and (λ_i, x_i) for $1 \leq i \leq k$ be eigenpairs of this matrix. Prove that if $\lambda_i \neq \lambda_j$ when $i \neq j$ then the eigenvectors x_i are linearly independent. In other words, given a set of distinct eigenvalues, a set of vectors created by taking one eigenvector per eigenvalue is linearly independent.

☒ done



Solution

For a complete solution, see the [notes](#).

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Homework 9.2.1.7

1/1 point (graded)
Consider the matrices

$$A = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}$$

and

$$\begin{pmatrix} \begin{array}{cccc|ccc} 4 & -1 & & & -1 & & & \\ -1 & 4 & -1 & & & -1 & & \\ & -1 & 4 & -1 & & & -1 & \\ & & -1 & 4 & & & & -1 \\ \hline -1 & & & & 4 & -1 & & -1 \\ & -1 & & & -1 & 4 & -1 & \ddots \\ & & -1 & & & -1 & 4 & -1 \\ & & & -1 & & & -1 & 4 \\ \hline & & & & -1 & & & 4 \end{array} & \begin{array}{ccc|cc} & & & \ddots & \\ & & & & \ddots \\ & & & & & \ddots \\ & & & & & & \ddots \\ \hline & & & & & & & \ddots \end{array} \end{pmatrix}$$

ALWAYS/SOMETIMES/NEVER: All eigenvalues of these matrices are nonnegative.

ALWAYS

 Answer: ALWAYS

ALWAYS/SOMETIMES/NEVER: All eigenvalues of the first matrix are positive.



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